

WGUPS Routing Program

Evidence Screenshots

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The following figures show the program output at the times required by the scenario.

Status Check at 8:35 a.m.

Figure 1. Package status at 8:35 a.m. All forty packages are visible on their assigned trucks. Packages 6, 9, 25, 28, and 32 appear in “delayed” status. Packages 2, 7, 14, 15, 16, 29, 33, and 34 are already delivered. Truck 1 is en route. Truck 2 packages remain at the hub because that truck does not depart until 9:05 a.m. Truck 3 packages also remain at the hub because that truck has not yet departed.

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bradwhite /c950 -> just run-compiled
WGUPS Routing Program
Total mileage for all trucks: 89.3 miles
Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: 8:35 AM
Status at 8:35 AM:
Package 1: at hub (Truck 2, Deadline: 10:30:00)
Package 2: delivered at 8:34 AM (Truck 1, Deadline: EOD)
Package 3: at hub (Truck 2, Deadline: EOD)
Package 4: at hub (Truck 2, Deadline: EOD)
Package 5: at hub (Truck 3, Deadline: EOD)
Package 6: delayed (Truck 2, Deadline: 10:30:00)
Package 7: delivered at 8:29 AM (Truck 1, Deadline: EOD)
Package 8: at hub (Truck 3, Deadline: EOD)
Package 9: delayed (Truck 3, Deadline: EOD)
Package 10: at hub (Truck 2, Deadline: EOD)
Package 11: at hub (Truck 2, Deadline: EOD)
Package 12: en route (Truck 1, Deadline: EOD)
Package 13: en route (Truck 1, Deadline: 10:30:00)
Package 14: delivered at 8:18 AM (Truck 1, Deadline: 10:30:00)
Package 15: delivered at 8:11 AM (Truck 1, Deadline: 09:00:00)
Package 16: delivered at 8:11 AM (Truck 1, Deadline: 10:30:00)
Package 17: at hub (Truck 2, Deadline: EOD)
Package 18: at hub (Truck 2, Deadline: EOD)
Package 19: en route (Truck 1, Deadline: EOD)
Package 20: en route (Truck 1, Deadline: 10:30:00)
Package 21: en route (Truck 1, Deadline: EOD)
Package 22: at hub (Truck 3, Deadline: EOD)
Package 23: at hub (Truck 2, Deadline: EOD)
Package 24: at hub (Truck 3, Deadline: EOD)
Package 25: delayed (Truck 2, Deadline: 10:30:00)
Package 26: at hub (Truck 3, Deadline: EOD)
Package 27: at hub (Truck 3, Deadline: EOD)
Package 28: delayed (Truck 2, Deadline: EOD)
Package 29: delivered at 8:29 AM (Truck 1, Deadline: 10:30:00)
Package 30: at hub (Truck 2, Deadline: 10:30:00)
Package 31: en route (Truck 1, Deadline: 10:30:00)
Package 32: delayed (Truck 2, Deadline: EOD)
Package 33: delivered at 8:34 AM (Truck 1, Deadline: EOD)
Package 34: delivered at 8:11 AM (Truck 1, Deadline: 10:30:00)
Package 35: at hub (Truck 3, Deadline: EOD)
Package 36: at hub (Truck 2, Deadline: EOD)
Package 37: at hub (Truck 2, Deadline: 10:30:00)
Package 38: at hub (Truck 2, Deadline: EOD)
Package 39: en route (Truck 1, Deadline: EOD)
Package 40: en route (Truck 1, Deadline: 10:30:00)
Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit:
```

Status Check at 9:35 a.m.

Figure 2. Package status at 9:35 a.m. Package 9 is still “delayed” because its address is not corrected until 10:20 a.m. Packages 6, 25, 28, and 32 are now “en route” or already delivered because Truck 2 departed at 9:05 a.m. Truck 3’s packages remain “at hub” pending the 10:30 a.m. departure.

```
Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: 9:35 AM

Status at 9:35 AM:

Package 1: en route (Truck 2, Deadline: 10:30:00)
Package 2: delivered at 8:34 AM (Truck 1, Deadline: E00)
Package 3: en route (Truck 2, Deadline: E00)
Package 4: en route (Truck 2, Deadline: E00)
Package 5: at hub (Truck 3, Deadline: E00)
Package 6: en route (Truck 2, Deadline: 10:30:00)
Package 7: delivered at 8:29 AM (Truck 1, Deadline: E00)
Package 8: at hub (Truck 3, Deadline: E00)
Package 9: delayed (Truck 3, Deadline: E00)
Package 10: en route (Truck 2, Deadline: E00)
Package 11: delivered at 9:29 AM (Truck 2, Deadline: E00)
Package 12: delivered at 9:29 AM (Truck 1, Deadline: E00)
Package 13: delivered at 9:05 AM (Truck 1, Deadline: 10:30:00)
Package 14: delivered at 8:18 AM (Truck 1, Deadline: 10:30:00)
Package 15: delivered at 8:11 AM (Truck 1, Deadline: 09:00:00)
Package 16: delivered at 8:11 AM (Truck 1, Deadline: 10:30:00)
Package 17: en route (Truck 2, Deadline: E00)
Package 18: delivered at 9:33 AM (Truck 2, Deadline: E00)
Package 19: delivered at 9:34 AM (Truck 1, Deadline: E00)
Package 20: en route (Truck 1, Deadline: 10:30:00)
Package 21: en route (Truck 1, Deadline: E00)
Package 22: at hub (Truck 3, Deadline: E00)
Package 23: delivered at 9:31 AM (Truck 2, Deadline: E00)
Package 24: at hub (Truck 3, Deadline: E00)
Package 25: delivered at 9:13 AM (Truck 2, Deadline: 10:30:00)
Package 26: at hub (Truck 3, Deadline: E00)
Package 27: at hub (Truck 3, Deadline: E00)
Package 28: en route (Truck 2, Deadline: E00)
Package 29: delivered at 8:29 AM (Truck 1, Deadline: 10:30:00)
Package 30: en route (Truck 2, Deadline: 10:30:00)
Package 31: delivered at 8:46 AM (Truck 1, Deadline: 10:30:00)
Package 32: en route (Truck 2, Deadline: E00)
Package 33: delivered at 8:34 AM (Truck 1, Deadline: E00)
Package 34: delivered at 8:11 AM (Truck 1, Deadline: 10:30:00)
Package 35: at hub (Truck 3, Deadline: E00)
Package 36: en route (Truck 2, Deadline: E00)
Package 37: en route (Truck 2, Deadline: 10:30:00)
Package 38: en route (Truck 2, Deadline: E00)
Package 39: delivered at 9:05 AM (Truck 1, Deadline: E00)
Package 40: delivered at 8:40 AM (Truck 1, Deadline: 10:30:00)

Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: █
```

Status Check at 12:03 p.m.

Figure 3. Package status at 12:03 p.m. All forty packages show “delivered at “ with no pending deliveries. Truck 3’s last deliveries occurred around 11:10 a.m. to 11:46 a.m., confirming that the day ends only when every package has been served.

```
Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: 12:03 PM
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Status at 12:03 PM:

```
Package 1: delivered at 10:09 AM      (Truck 2, Deadline: 10:30:00)
Package 2: delivered at 8:34 AM       (Truck 1, Deadline: EOD)
Package 3: delivered at 10:26 AM      (Truck 2, Deadline: EOD)
Package 4: delivered at 10:03 AM      (Truck 2, Deadline: EOD)
Package 5: delivered at 11:10 AM      (Truck 3, Deadline: EOD)
Package 6: delivered at 9:51 AM       (Truck 2, Deadline: 10:30:00)
Package 7: delivered at 8:29 AM       (Truck 1, Deadline: EOD)
Package 8: delivered at 11:10 AM      (Truck 3, Deadline: EOD)
Package 9: delivered at 11:13 AM      (Truck 3, Deadline: EOD)
Package 10: delivered at 10:36 AM     (Truck 2, Deadline: EOD)
Package 11: delivered at 9:29 AM      (Truck 2, Deadline: EOD)
Package 12: delivered at 9:29 AM      (Truck 1, Deadline: EOD)
Package 13: delivered at 9:05 AM      (Truck 1, Deadline: 10:30:00)
Package 14: delivered at 8:18 AM      (Truck 1, Deadline: 10:30:00)
Package 15: delivered at 8:11 AM      (Truck 1, Deadline: 09:00:00)
Package 16: delivered at 8:11 AM      (Truck 1, Deadline: 10:30:00)
Package 17: delivered at 9:56 AM      (Truck 2, Deadline: EOD)
Package 18: delivered at 9:33 AM      (Truck 2, Deadline: EOD)
Package 19: delivered at 9:34 AM      (Truck 1, Deadline: EOD)
Package 20: delivered at 9:36 AM      (Truck 1, Deadline: 10:30:00)
Package 21: delivered at 9:36 AM      (Truck 1, Deadline: EOD)
Package 22: delivered at 11:46 AM     (Truck 3, Deadline: EOD)
Package 23: delivered at 9:31 AM      (Truck 2, Deadline: EOD)
Package 24: delivered at 11:36 AM     (Truck 3, Deadline: EOD)
Package 25: delivered at 9:13 AM      (Truck 2, Deadline: 10:30:00)
Package 26: delivered at 11:42 AM     (Truck 3, Deadline: EOD)
Package 27: delivered at 10:54 AM     (Truck 3, Deadline: EOD)
Package 28: delivered at 10:07 AM     (Truck 2, Deadline: EOD)
Package 29: delivered at 8:29 AM      (Truck 1, Deadline: 10:30:00)
Package 30: delivered at 10:24 AM     (Truck 2, Deadline: 10:30:00)
Package 31: delivered at 8:46 AM      (Truck 1, Deadline: 10:30:00)
Package 32: delivered at 9:58 AM      (Truck 2, Deadline: EOD)
Package 33: delivered at 8:34 AM      (Truck 1, Deadline: EOD)
Package 34: delivered at 8:11 AM      (Truck 1, Deadline: 10:30:00)
Package 35: delivered at 10:54 AM     (Truck 3, Deadline: EOD)
Package 36: delivered at 9:46 AM      (Truck 2, Deadline: EOD)
Package 37: delivered at 10:24 AM     (Truck 2, Deadline: 10:30:00)
Package 38: delivered at 10:24 AM     (Truck 2, Deadline: EOD)
Package 39: delivered at 9:05 AM      (Truck 1, Deadline: EOD)
Package 40: delivered at 8:40 AM      (Truck 1, Deadline: 10:30:00)
```

```
Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit:
```

Total Mileage

Figure 4. Total mileage after complete execution. The output header reads “Total mileage for all trucks: 89.3 miles,” which is below the 140-mile requirement. Every package shows its final delivered status, truck assignment, and timestamp.

```
Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: all
Package 1: delivered at 10:09 AM (Truck 2, 195 W Oakland Ave)
Package 2: delivered at 8:34 AM (Truck 1, 2530 S 500 E)
Package 3: delivered at 10:26 AM (Truck 2, 233 Canyon Rd)
Package 4: delivered at 10:03 AM (Truck 2, 380 W 2880 S)
Package 5: delivered at 11:10 AM (Truck 3, 300 State St)
Package 6: delivered at 9:51 AM (Truck 2, 3060 Lester St)
Package 7: delivered at 8:29 AM (Truck 1, 1330 2100 S)
Package 8: delivered at 11:10 AM (Truck 3, 300 State St)
Package 9: delivered at 11:13 AM (Truck 3, 410 S State St)
Package 10: delivered at 10:36 AM (Truck 2, 600 E 900 South)
Package 11: delivered at 9:29 AM (Truck 2, 2600 Taylorsville Blvd)
Package 12: delivered at 9:29 AM (Truck 1, 3575 W Valley Central Station bus Loop)
Package 13: delivered at 9:05 AM (Truck 1, 2010 W 500 S)
Package 14: delivered at 8:18 AM (Truck 1, 4300 S 1300 E)
Package 15: delivered at 8:11 AM (Truck 1, 4580 S 2300 E)
Package 16: delivered at 8:11 AM (Truck 1, 4580 S 2300 E)
Package 17: delivered at 9:56 AM (Truck 2, 3148 S 1100 W)
Package 18: delivered at 9:33 AM (Truck 2, 1488 4800 S)
Package 19: delivered at 9:34 AM (Truck 1, 177 W Price Ave)
Package 20: delivered at 9:36 AM (Truck 1, 3595 Main St)
Package 21: delivered at 9:36 AM (Truck 1, 3595 Main St)
Package 22: delivered at 11:46 AM (Truck 3, 6351 South 900 East)
Package 23: delivered at 9:31 AM (Truck 2, 5100 South 2700 West)
Package 24: delivered at 11:36 AM (Truck 3, 5025 State St)
Package 25: delivered at 9:13 AM (Truck 2, 5383 S 900 East #104)
Package 26: delivered at 11:42 AM (Truck 3, 5383 S 900 East #104)
Package 27: delivered at 10:54 AM (Truck 3, 1060 Dalton Ave S)
Package 28: delivered at 10:07 AM (Truck 2, 2835 Main St)
Package 29: delivered at 8:29 AM (Truck 1, 1330 2100 S)
Package 30: delivered at 10:24 AM (Truck 2, 300 State St)
Package 31: delivered at 8:46 AM (Truck 1, 3365 S 900 W)
Package 32: delivered at 9:58 AM (Truck 2, 3365 S 900 W)
Package 33: delivered at 8:34 AM (Truck 1, 2530 S 500 E)
Package 34: delivered at 8:11 AM (Truck 1, 4580 S 2300 E)
Package 35: delivered at 10:54 AM (Truck 3, 1060 Dalton Ave S)
Package 36: delivered at 9:46 AM (Truck 2, 2300 Parkway Blvd)
Package 37: delivered at 10:24 AM (Truck 2, 300 State St)
Package 38: delivered at 10:24 AM (Truck 2, 300 State St)
Package 39: delivered at 9:05 AM (Truck 1, 2010 W 500 S)
Package 40: delivered at 8:40 AM (Truck 1, 380 W 2880 S)
```

Total mileage: 89.3 miles

Enter a time (e.g. '8:35 AM', '12:03 PM', '1:12 PM') or 'all' for final status, or 'q' to quit: